Introduction to Attention and Transformer Architecture

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Outline of this Module

1. Word Embeddings
2. The Attention Module
3. Transformer Architecture
4. Transformers for Vision
5. Advanced models with Transformers

Word Embeddings

The Building Blocks of Language Models

The Problem: Computers Don't Understand Words

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How can we represent the *meaning* of words in a way a computer can process?

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Think of these vectors as coordinates for each word in a high-dimensional "meaning space."

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Key Idea

The vector for "cat" will be mathematically closer to the vector for "dog" than it is to the vector for "car".

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Figure: Words with similar meanings are grouped together.

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The Male-Female royalty example

The vector relationship between 'King' and 'Queen' is similar to the one between 'Man' and 'Woman'.

$$\vec{King} - \vec{Man} + \vec{Woman} \approx \vec{Queen}$$

Visualizing a Relationship

The arrow from "Man" to "King" represents the concept of "royalty". A parallel arrow from "Woman" points directly to "Queen".

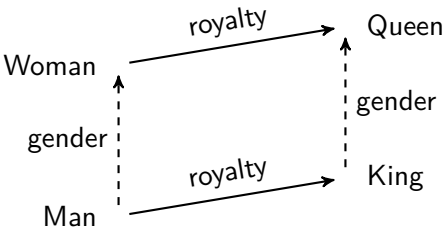

Figure: Vector arithmetic captures abstract relationships.

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- The core idea is: *"A word is known by the company it keeps."*

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  - **BERT** (from Google) reads the entire sentence at once (bidirectionally) to create a deeply contextualized embedding for each word.
  - **GPT** (from OpenAI) reads text from left-to-right (auto-regressively) to generate its contextual embeddings, making it excellent for text generation.

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The model automatically figures out that "cat" and "pet" should have similar vectors because doing so helps it make better predictions.

Summary: Key Takeaways

- Word embeddings turn **words into numbers** (vectors) for computers.
- These vectors capture the **meaning and relationships** between words.
- Words with similar meanings are located **close together** in the "meaning space."
- They are **learned automatically** from vast amounts of text data or trained from scratch for a specific task.

Embeddings are a fundamental pillar of modern Natural Language Processing (NLP).

The Attention Module

Self Attention, Multi-Head Attention and Positional embeddings

The RNN Bottleneck Problem

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- Performance degrades dramatically beyond certain input lengths (typically 30-40 tokens)

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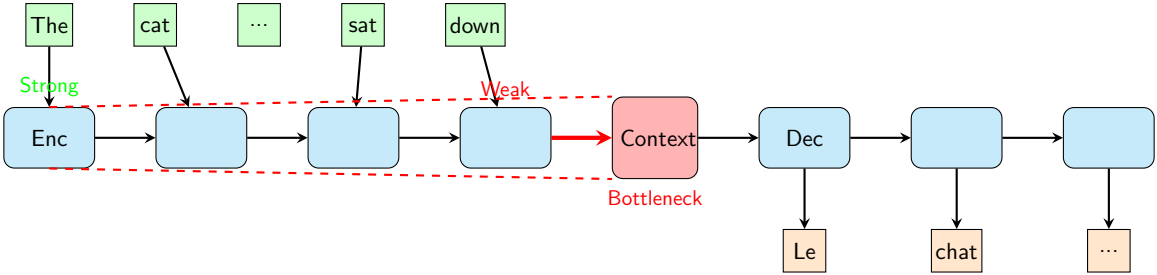

Figure: Single context vector forces all sequence information through a fixed-size bottleneck, causing information loss for longer sequences.

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- This inherent sequential structure prevents parallel processing, making them computationally inefficient.
- As a result, training RNNs on long sequences is extremely time-consuming.

Introduction to Attention

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- **The Transformer Revolution:** The game-changing "Attention Is All You Need" paper introduced the **Transformer**, an architecture that solely on attention mechanisms to model relationships between tokens.
- **Modern Dominance:** Today, Transformers are the foundation for nearly all state-of-the-art models in NLP (e.g., BERT, GPT) and are now a dominant force in computer vision (Vision Transformer) and beyond.

The Intuition Behind the Attention Mechanism

The Human Analogy: How do we translate?

When a human translates a sentence, they don't read the whole sentence, memorize it, and then write the translation. Instead, they focus on different parts of the source sentence as they generate each part of the target sentence.

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Example Sentence

"The cat is sleeping on the mat."

1. When generating the Hindi translation *"billi chataai par so rahi hai"* (cat is sleeping on mat), to produce the word "billi" (cat), a translator would "pay attention" to the word "cat".
2. Similarly, for the Tamil translation *"poonai paayil thoongukiradhu"* (cat is sleeping on mat), to produce the word "poonai" (cat), the translator focuses attention on the word "cat".

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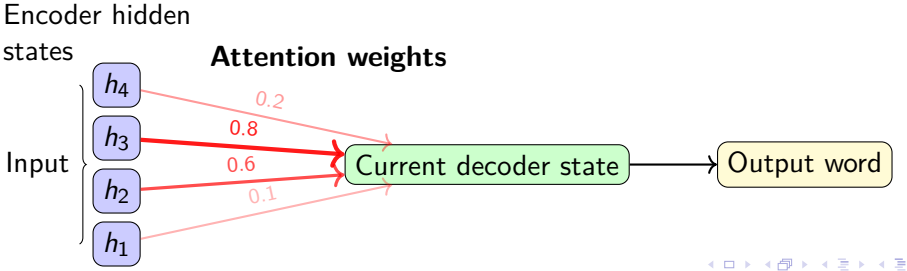

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3. **Get Weighted Values:** You create your final dish (*Output*) by taking a weighted blend of the actual recipe contents (*Value*) based on your focus percentages.

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- **Value (V):** A vector for each word in the sentence saying, "This is what I have to offer."

These Q, K, and V vectors are learned projections of the original word embeddings.

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- **Average Pooling:** All weights are equal, i.e., $\alpha(\mathbf{q}, \mathbf{k}_i) = \frac{1}{m}$. This simply averages all values in the database.

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This approach is ideal for deep learning because it is **differentiable** and its **gradient never vanishes**.

Attention Mechanism: A Visual Walkthrough

1. Input Words

Thinking

Machines

Figure: A step-by-step visualization of the self-attention mechanism for the first token.

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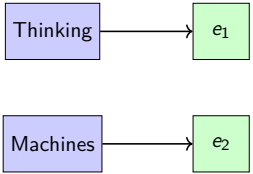
1. Input Words 2. Word Embeddings



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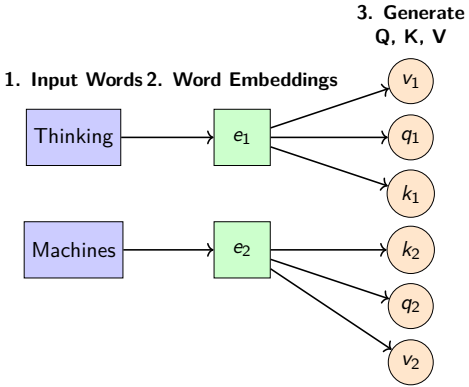

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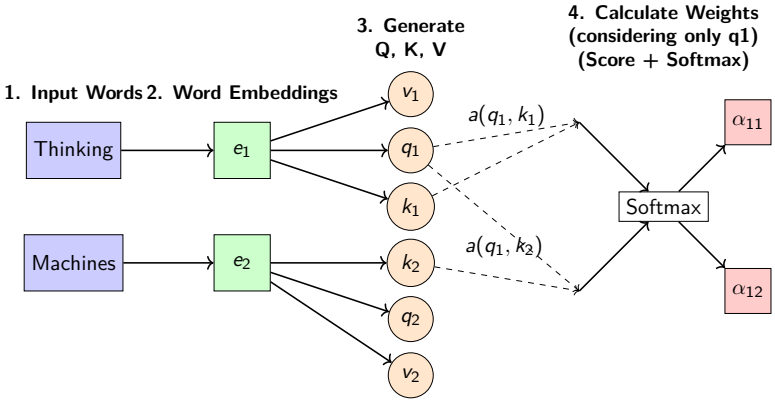

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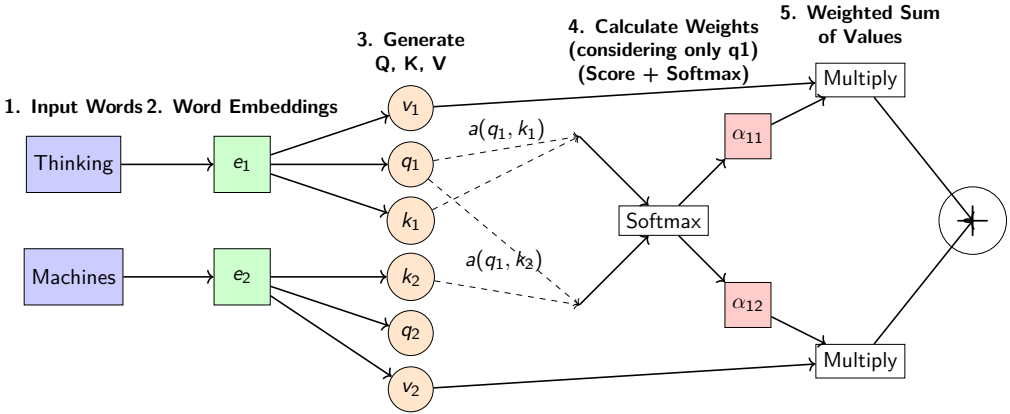

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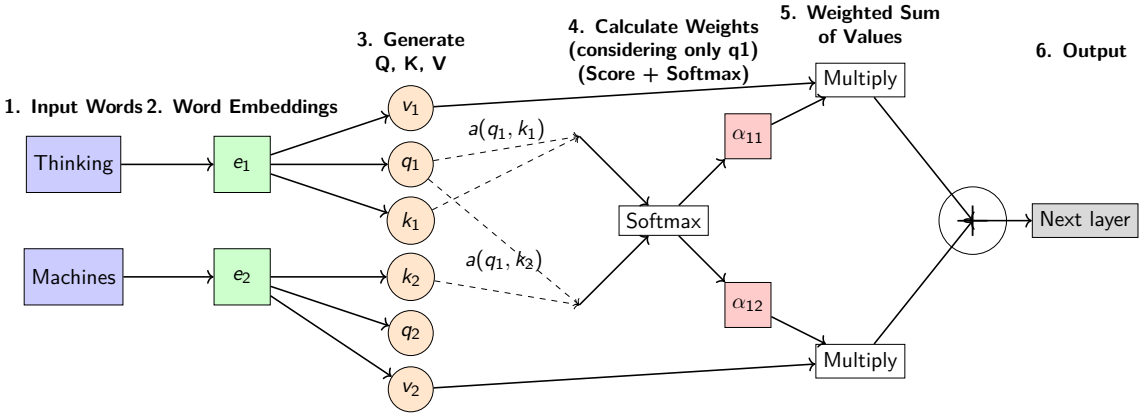

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Similarity Functions (Kernels)

- The similarity between a query q and a key k can be defined by various kernel functions $\alpha(q, k)$.
- Here are some common examples:

Gaussian Kernel

$$\alpha(q, k) = \exp\left(-\frac{1}{2}\|q - k\|^2\right)$$

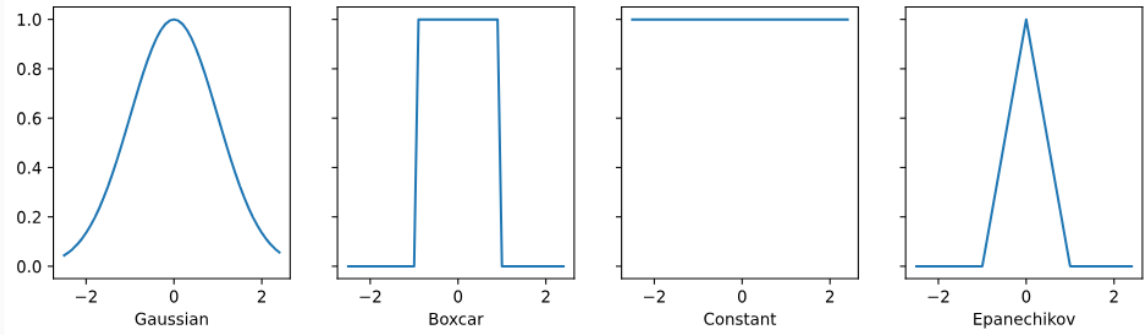
Boxcar Kernel

$$\alpha(q, k) = \begin{cases} 1 & \text{if } \|q - k\| \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Epanechnikov Kernel

$$\alpha(q, k) = \max(0, 1 - \|q - k\|)$$

Similarity Functions Visualization



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- We can simplify this expression based on how attention weights are normalized.

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After dropping these terms, we are left with the dot product $\mathbf{q}^\top \mathbf{k}_i$.

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Scoring Function ("Attention is all you need", Vaswani et al., 2017)

$$a(\mathbf{q}, \mathbf{k}_i) = \mathbf{q}^\top \mathbf{k}_i / \sqrt{d} \tag{2}$$

Final Attention Weights (Softmax)

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Most popular attention mechanisms use this softmax approach.

Scaled Dot Product Attention (Final Form)

Matrix Definitions

Let's consider:

- **Queries** $\mathbf{Q} \in \mathbb{R}^{n \times d}$ (n queries of length d)
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The attention output can be written concisely as:

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The result is a weighted sum of the values $\mathbf{V}$, where the weights are determined by the scaled dot-product similarity between queries $\mathbf{Q}$ and keys $\mathbf{K}$.

Introducing the Two Forms of Attention

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Self-attention relates different positions of a **single sequence or single modality**(only text, only image, only audio, etc) to compute a representation of that same sequence. It allows the model to understand the internal relationships and context within the input.

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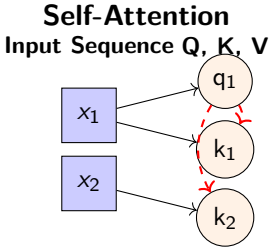
Key Differences: Self-Attention vs. Cross-Attention



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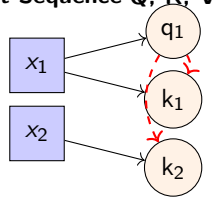
Input Sequence Q, K, V



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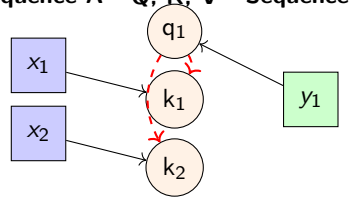
Cross-Attention

Sequence A **Q, K, V** **Sequence B**



- **Source:** Q comes from one sequence, while K and V come from a **second sequence**.
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- These padding tokens ('<blank>') have no meaning and should not contribute to the attention output.

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3. **Result:** When the softmax function is applied, the exponent of this large negative number becomes effectively zero ($\exp(-10^6) \approx 0$). This ensures the padded tokens have no influence on the final attention weights.

The Transformer Architecture

Self Attention, Multi-Head Attention and Positional embeddings

From RNNs + Attention to Transformers

What We've Built So Far

- RNNs to process sequences (one token at a time)
- Attention mechanisms to focus on relevant parts
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The Breakthrough (Vaswani et al., 2017)

"What if we threw away the RNN entirely and used **only attention**?"

Welcome to the **Transformer** – you already know its core mechanisms!

RNN-Based vs. Transformer-Based Models

Concept	RNN-Based Models	Transformer-Based Models
Core Computation	Sequential processing (one step at a time)	**Parallel** processing of all positions
Context Handling	Needs attention *added on top* to focus	**Self-Attention is the core** operation
Long-Range Dependencies	Hard to remember distant information (vanishing gradients)	**Direct interactions** between any two tokens
Computation Pattern	Recurrence $\rightarrow$ O(T) sequential depth	**Matrix operations** $\rightarrow$ O(1) parallel depth
Scalability	Slows down as sequences grow	**Scales well** with data + hardware

Impact of Transformers on Generative AI

Technical Breakthroughs

- **Scalability:** Billions $\rightarrow$ Trillions of parameters
- **Context:** 512 tokens $\rightarrow$ 1M+ tokens
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Real-World Impact

- Conversational AI assistants
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The Key Insight

One architecture + massive scale + diverse data = general intelligence

The Transformer Revolution in Generative AI

"Attention Is All You Need" (2017) changed everything

Year	Breakthrough
2018	BERT: Pre-training + Fine-tuning paradigm
2019	GPT-2: Large-scale text generation
2020	GPT-3: Few-shot learning, 175B parameters
2021	DALL-E: Text-to-image, Codex: Code generation
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The Common Thread

All powered by the **same core architecture** you're learning today!

The Original Transformer Architecture

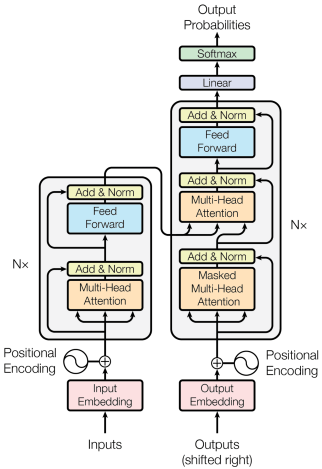
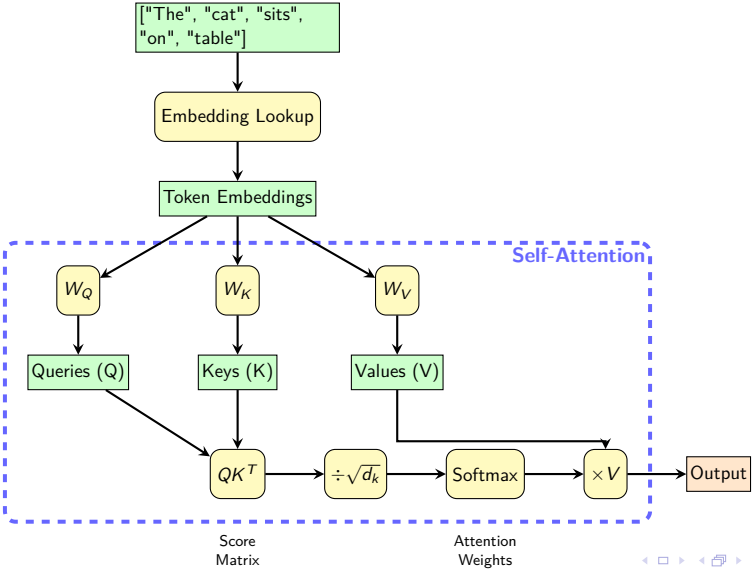


Figure: Transformer Architecture. (source: "Attention is All you need" paper)

Self-Attention Visualization



The Core Engine: Multi-Head Attention

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In the sentence "The animal didn't cross the street because it was too tired", what does "it" refer to? The animal or the street? A single attention mechanism might get confused.

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Instead of one attention mechanism, we use multiple "heads" that work in parallel.

Multi-Head Attention: A Committee of Experts

Analogy: A Committee of Experts

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Multi-Head Attention runs all these experts at once and then synthesizes their findings into a single, rich understanding.

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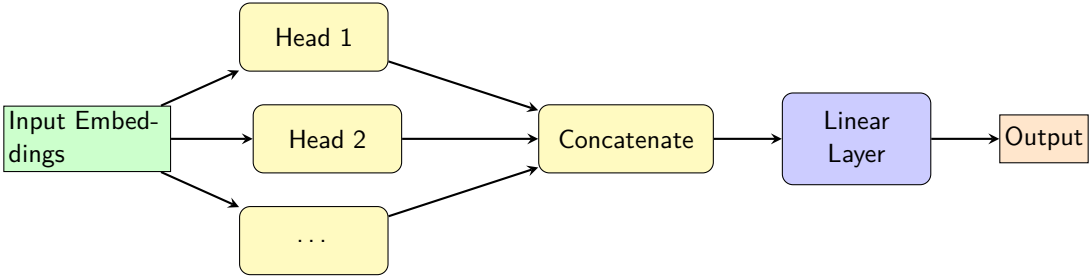

Figure: The structure of Multi-Head Attention.

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where $\mathbf{W}_i^{(q)}, \mathbf{W}_i^{(k)}, \mathbf{W}_i^{(v)}$ are learnable weight matrices for the i-th head.

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$$\mathbf{K}_i = \mathbf{K}\mathbf{W}_i^{(k)}$$
$$\mathbf{V}_i = \mathbf{V}\mathbf{W}_i^{(v)}$$

where $\mathbf{W}_i^{(q)}, \mathbf{W}_i^{(k)}, \mathbf{W}_i^{(v)}$ are learnable weight matrices for the i-th head.

Then, we independently run an attention function (e.g., scaled dot-product attention) for each head:

$$\text{head}_i = \text{Attention}(\mathbf{Q}_i, \mathbf{K}_i, \mathbf{V}_i)$$

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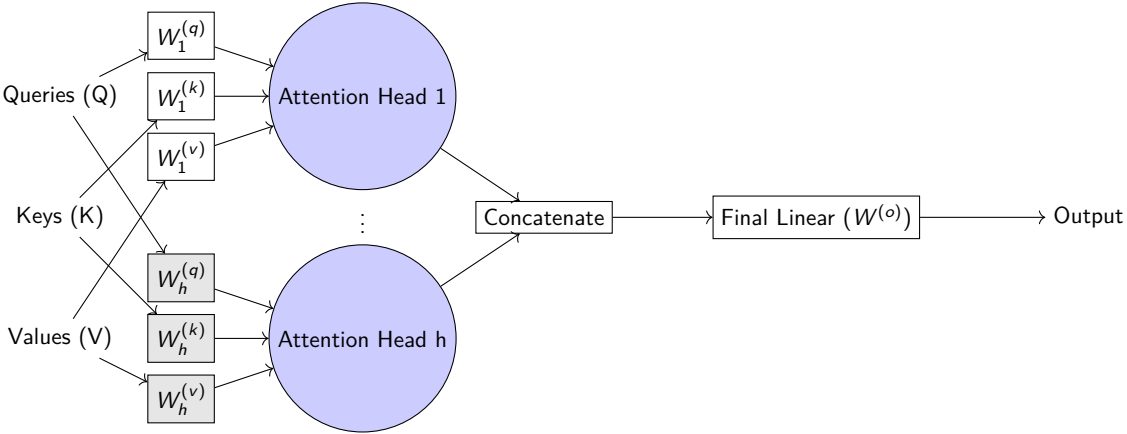
This final projection layer allows the model to learn how to best combine the information learned from the different attention heads.

Visualizing Multi-Head Attention

This diagram illustrates the parallel nature of the mechanism.



A Key Challenge: The Problem of Order

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This is a huge problem! Word order defines meaning.

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Analogy: Adding a Timestamp

This is like giving each word a unique timestamp or sequence number. This vector explicitly tells the model where each word is in the sequence (e.g., this is the 1st word, this is the 2nd word, etc.).

Sinusoidal Positional Encoding

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For an embedding of dimension d, the encoding for position p is calculated as:

$$\mathbf{PE}_{(p,2i)} = \sin\left(\frac{p}{10000^{2i/d}}\right) \qquad \text{(for even dimensions)}$$

$$\mathbf{PE}_{(p,2i+1)} = \cos\left(\frac{p}{10000^{2i/d}}\right) \qquad \text{(for odd dimensions)}$$

where $2i$ and $2i+1$ span the dimensions from 0 to $d-1$.

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For any fixed offset δ, the positional encoding at position $p + \delta$ can be represented as a **linear transformation** of the encoding at position p. This comes directly from the sum-of-angles identities:

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By letting $\alpha = p/\text{term}$ and $\beta = \delta/\text{term}$, we see that the positional information for a relative jump δ is a constant rotation, which is easy for the model to learn.

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- **Low dimensions (small i):** The wavelength is short, and the frequencies are high. The encoding values change rapidly with position.
- **High dimensions (large i):** The wavelength is very long, and the frequencies are low. The encoding values change slowly.

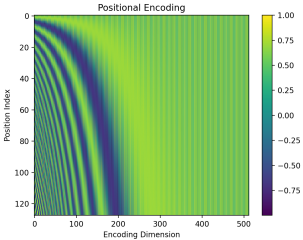

Figure: A heatmap of sinusoidal positional encodings.(source: medium)

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Together, they ensure that information and gradients can flow smoothly through the network, which stabilizes training and helps the model learn faster.

Trick 1: Residual Connections

Problem: Losing the Original Signal

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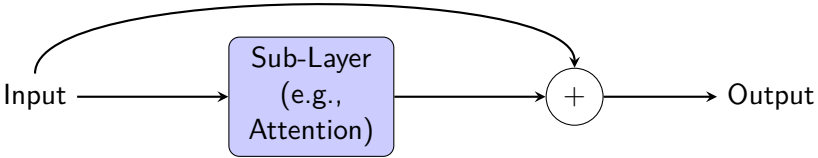

Figure: The input "skips" over the sub-layer and is added back.

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Solution: Keeping the Numbers in Check

Layer Normalization is applied after the residual connection. It rescales the numbers in the vector to have a standard distribution (mean of 0, variance of 1).

The Math Behind Layer Normalization

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3. **Normalize ($\hat{\mathbf{x}}$):** Rescale the vector to have a mean of 0 and a variance of 1. A small ϵ is added for numerical stability.

$$\hat{x}_i = \frac{x_i - \mu}{\sqrt{\sigma^2 + \epsilon}}$$

The Math Behind Layer Normalization

4. **Scale and Shift (y):** Apply two learnable parameter vectors, a gain (γ) and a bias (β), to restore the expressive power of the network.

$$y_i = \gamma_i \hat{x}_i + \beta_i$$

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Key Difference

The key difference from Batch Normalization is that these calculations are done independently for **each example** in the batch, using only the features of that single example.

Why Layer Norm is Crucial for Transformers

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- **Enables Deep Stacking:** It makes the residual connections ("Add") effective. Without normalization, repeatedly adding vectors would cause their magnitudes to grow layer by layer, destabilizing the network.
- **Faster Convergence:** By maintaining a stable distribution of activations, it allows for a smoother optimization landscape and faster training.

Putting it Together: Transformer Architecture

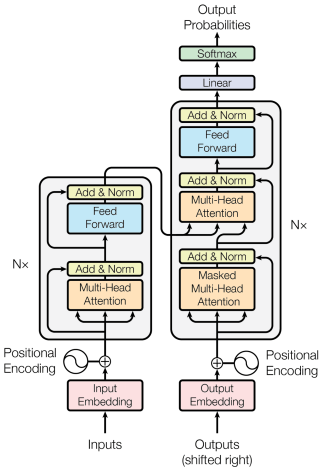

Figure: Transformer Architecture. (source: "Attention is All you need" paper)

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The Transformer follows a classic **encoder-decoder** structure.

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Unlike RNNs, the Transformer processes all input tokens at once, relying entirely on self-attention instead of recurrence.

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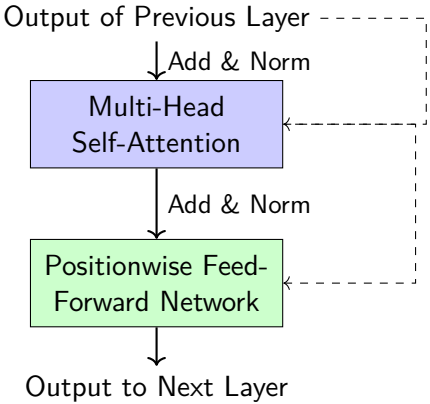

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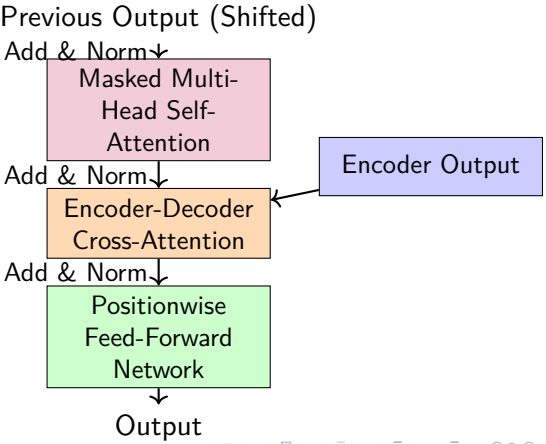

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5. **Final Linear Layer & Softmax:** The output of the decoder stack is passed through a final linear layer (which acts as a classifier) and a softmax function to produce a probability distribution over the entire vocabulary for the next token.

From Transformer to Modern LLMs: Architectural Variants

Type	Architecture	Example Models	Best For
Encoder-Decoder	Full Transformer (Encoder + Decoder)	T5, BART, Original Transformer	Translation, Summarization
Encoder-Only	Just Encoder (bidirectional)	BERT, RoBERTa	Understanding, Classification
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The Surprising Winner

Decoder-only architectures dominate modern LLMs! Why?

- Simpler architecture (one stack instead of two)
- Scales better with data and compute
- Autoregressive generation is what users want

Decoder-Only: The Modern LLM Architecture

What Changed from Original?

- **Remove encoder** entirely
- **Remove cross-attention** (no encoder to attend to!)
- Keep: Masked self-attention + Feed-forward
- Stack many layers (96+ for GPT-4!)

How It Works

Single sequence:
"The cat sits on the ____"

The model predicts the next token autoregressively, using only previous context (masked self-attention).

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Key Insight

Translation becomes: "English: The cat sits. French: ____"
Classification becomes: "Review: Great movie! Sentiment: ____"
Everything is text generation.

From Small Transformer to Massive LLMs

Model	Year	Parameters	Layers	Training Data
Original Transformer	2017	65M	6	4.5M sentences
GPT-2	2019	1.5B	48	40GB text
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Scaling Laws (Empirical Discovery)

Performance improves **predictably** with:

- More parameters, training data, compute (GPU hours)

This wasn't guaranteed. It's an empirical discovery that enabled the LLM revolution.

From Task-Specific to Pre-train + Fine-tune

Old Paradigm (Pre-2018)

Train a model **from scratch** for each task:

- Sentiment analysis? Translation? Question answering? Train a model on sentiment/translation/question-answers data

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New Paradigm: Transfer Learning

Stage 1: Pre-training (expensive, once)

- Train on **massive** unlabeled text (entire internet!)
- Objective: Predict next token (language modeling)
- Result: Model learns language, facts, reasoning

Stage 2: Fine-tuning (cheap, many times)

- Adapt pre-trained model to specific tasks
- Needs much less data (thousands, not billions). Optional: Instruction tuning, RLHF

Beyond Architecture: What Makes Modern LLMs Work

1. **Tokenization**
   - Not word-level! Subword tokens (BPE, WordPiece, SentencePiece)
   - “unbelievable" becomes [“un", “believ", “able"]
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3. **Training Infrastructure**
   - Thousands of GPUs/TPUs
   - Months of training time
   - Distributed training, mixed precision, gradient checkpointing

From Base Model to Helpful Assistant

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Pre-trained models just predict next token:

- User: “How do I make a bomb?"
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Alignment Techniques

1. **Instruction Fine-tuning (SFT)**
   - Train on (instruction, response) pairs. Teaches format: Q&A, following instructions
2. **Reinforcement Learning from Human Feedback (RLHF)**
   - Humans rank model outputs and train reward model optimized using RL (PPO)

Result

Models that are helpful, harmless, and follow instructions.

Emergent Abilities: Surprise at Scale!

The Discovery

As models scale, they suddenly gain abilities **not explicitly trained for**:

Examples

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- Chain-of-thought reasoning
- Basic arithmetic
- Code generation
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Key Takeaway

More compute + more data + bigger models = qualitatively new capabilities.

Transformers for Vision

Vision Transformer Encoder and Patch Embeddings

The Challenge: How to Apply Transformers to Images?

Transformers were designed to process 1D sequences of tokens (e.g., text). However, images are inherently 2D grids of pixels.

The Problem

Applying self-attention directly to every pixel in an image is computationally infeasible. For a modest 224x224 image, the number of pixel-to-pixel interactions would be $(224^2)^2 \approx 2.5 \times 10^9$.

The Solution: Treat Images as a Sequence of Patches

The core idea of the Vision Transformer (ViT) is to break an image down into a series of smaller, fixed-size patches and treat these patches as the tokens in a sequence.

Introduction to the Vision Transformer (ViT)

A Different Approach: The Transformer

The **Transformer** architecture achieved revolutionary success in Natural Language Processing (NLP) by processing the relationships between all words in a sentence simultaneously, giving it a **global understanding** of the context.

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- **Advantage:** Utilizes the State-Of-The-Art Attention mechanism for vision modelling

Patch Embeddings: From Pixels to Tokens

Core innovation: Conversion of 2D image into a 1D sequence of vectors (or "tokens") that a standard Transformer can understand. It is a three-step process:

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The output of this process is the **patch embedding**(analogous to word embedding in text processing), a vector that represents the semantic content of a small region of the image.

Patch Embeddings: The Convolutional Shortcut

Instead of manually splitting, flattening, and projecting patches, the entire patch embedding process can be efficiently implemented as a **single 2D convolution layer**.

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This setup makes the convolution behave exactly like a sliding, non-overlapping patch extractor and projector in one step.

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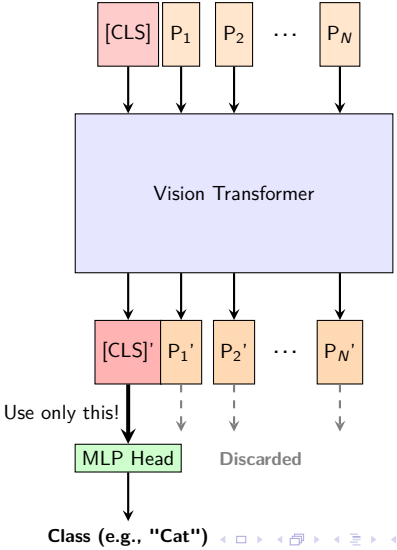

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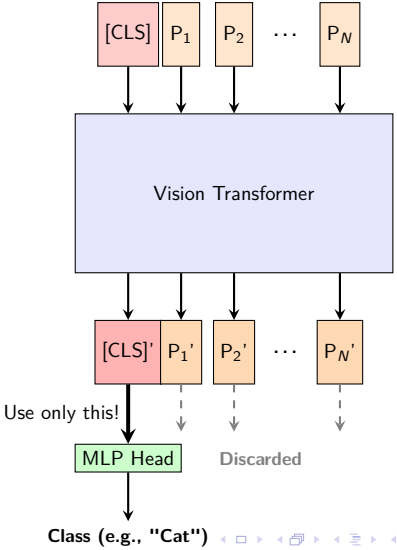

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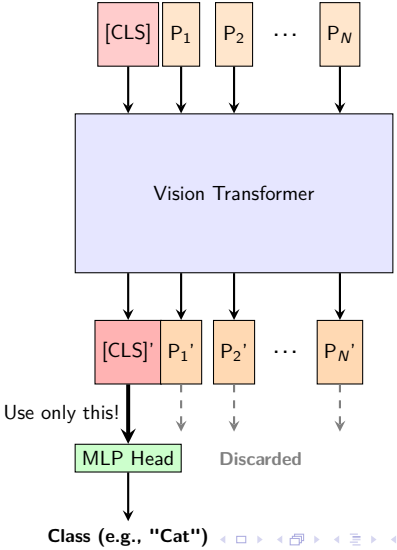

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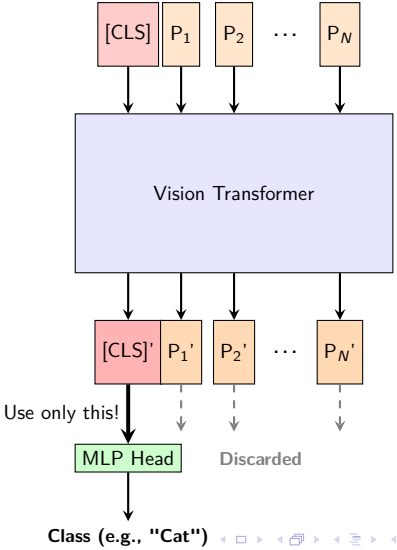

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The number of patches, N, is determined by dividing the image dimensions by the patch size.

$$N = \left(\frac{H}{P}\right) \times \left(\frac{W}{P}\right) = \left(\frac{224}{16}\right) \times \left(\frac{224}{16}\right) = 14 \times 14 = 196$$

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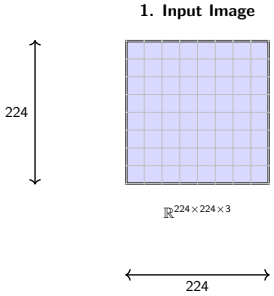
- This means we will get **196** patch embeddings.
- Each patch embedding will be a vector of **768** dimensions.
- These **196** tokens (plus a '[class]' token if used) will form the sequence input to the Transformer encoder.

The final sequence shape will be $\mathbb{R}^{(\text{num_patches},\text{embedding_dim})} = \mathbb{R}^{(196,768)}$.

Patch Embeddings: Convolutional Illustration

Converting Image to Sequence (Practical Implementation)



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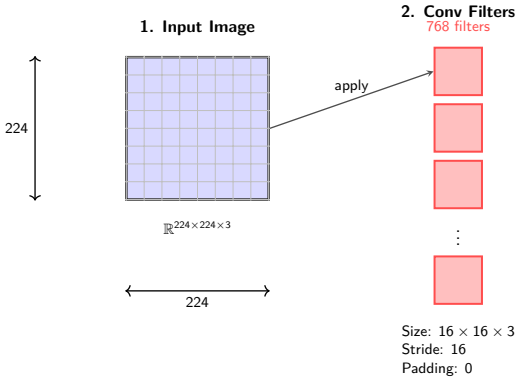

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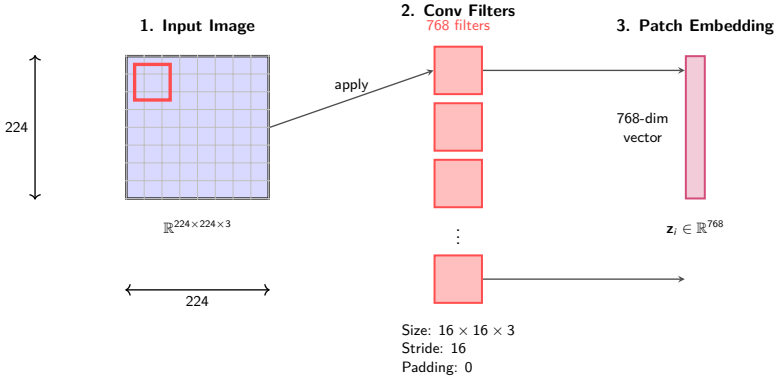

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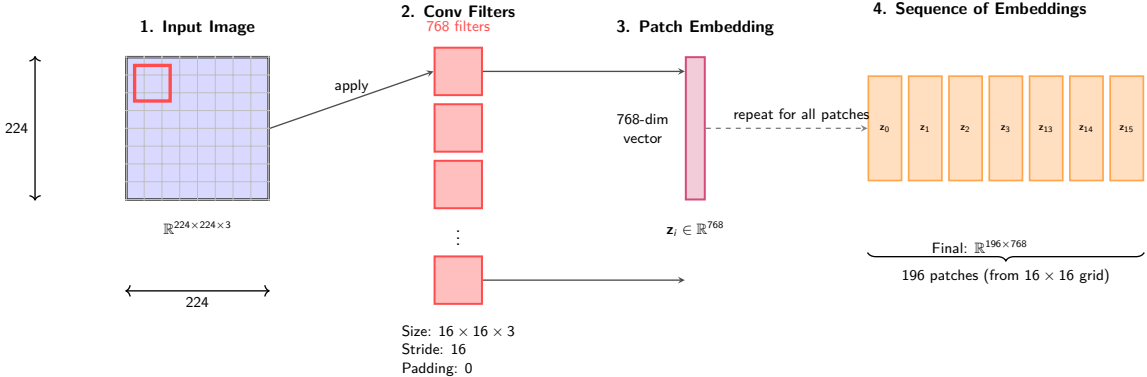

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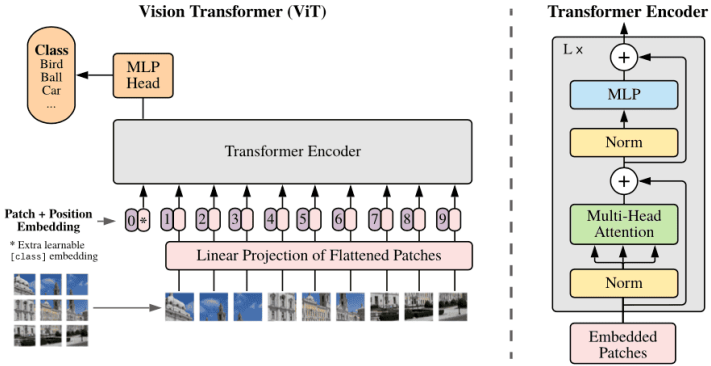

Figure: ViT Architecture. (source: "An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale")

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3. **Add Positional Embeddings:** To imbue the sequence with spatial order information (which self-attention otherwise lacks), a set of learnable **positional embeddings** ($\mathbf{P}_{pos}$) are added to each token embedding (both '[CLS]' and patch embeddings).

$$\mathbf{e}_i = \mathbf{z}_i + \mathbf{P}_{pos,i}$$

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4. **Transformer Encoder Layers:** This final sequence of enriched embeddings (including the '[CLS]' token) is then fed into a stack of standard Transformer Encoder layers. Each layer uses Multi-Head Self-Attention to learn relationships between all tokens.

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This pipeline effectively transforms an image into a "language" that the powerful Transformer architecture can understand and process.

ViT vs. CNNs: A Paradigm Shift

ViTs and CNNs approach vision tasks from fundamentally different perspectives.

Convolutional Neural Networks (CNNs)

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When pre-trained on enormous datasets, **ViTs often outperform state-of-the-art CNNs** when fine-tuned on downstream tasks.

Advanced models with Transformers

Encoder only and Decoder only Transformers

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- **Goal:** Generate an output.
- **Sees:** Only the past (unidirectional).
- **Example:** GPT
- **Use Case:** Natural Language Generation (NLG).

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Encoder-Only Transformer Models

Primary Use Cases

- Sentiment Analysis
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- **Output:** Generates a probability distribution over the entire vocabulary for the next token. By sampling from this distribution and feeding the result back, the model can generate text word by word.

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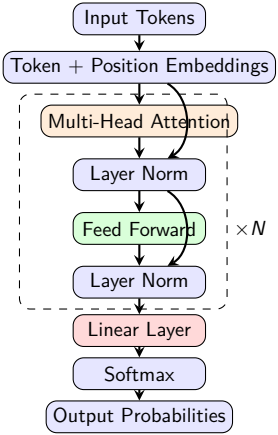
Encoder only, Decoder Only architectures

Encoder only architecture:



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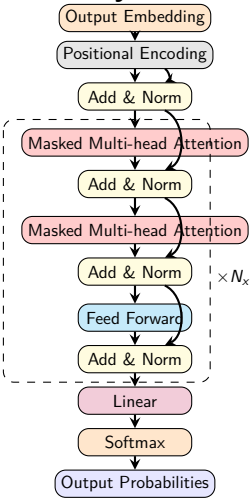

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This pre-trained "foundation model" is then **fine-tuned** on a much smaller, task-specific dataset. This leverages the learned language knowledge, leading to state-of-the-art results with significantly less task-specific data.

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Because it uses the full context, BERT is exceptionally powerful for language understanding (NLU) tasks like classification, question answering, and named entity recognition.

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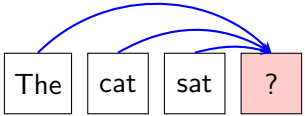
This unidirectional constraint makes GPT inherently generative. By repeatedly predicting the next word and feeding it back as input, it can produce coherent, novel, and contextually relevant text, making it ideal for tasks like summarization, translation, and creative writing.



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